

Student Registration System

Full Stack Development – TASK_1

1 Introduction

The Student Registration System is a web-based application developed using HTML5, CSS3, Bootstrap, Node.js, Express.js, and MySQL. It allows users to register student details, store them in a MySQL database, and retrieve them using SELECT queries.

2 Software Requirements

- Node.js
- MySQL Server
- MySQL Workbench
- VS Code
- Web Browser
- Git (Optional)

3 Steps Before Starting the Project

1. Install Node.js from official website.
2. Install MySQL Server and MySQL Workbench.
3. Create a new folder for project (TASK_1).
4. Open folder in VS Code.
5. Initialize Node project using command:

```
npm init -y
```

6. Install required packages:

```
npm install express mysql2 body-parser ejs dotenv
```

4 Project Structure

TASK_1/

```
server.js
package.json
.env
```

```
public/
  style.css
```

```
views/
  register.ejs
  students.ejs
```

5 Database Design

Database Name: studentdb

Table Name: students

5.1 SQL Code

```
1 CREATE DATABASE studentdb;
2
3 USE studentdb;
4
5 CREATE TABLE students (
6   id INT PRIMARY KEY AUTO_INCREMENT,
7   name VARCHAR(100) NOT NULL,
8   email VARCHAR(100) NOT NULL UNIQUE,
9   dob DATE NOT NULL,
10  department VARCHAR(100) NOT NULL,
11  phone VARCHAR(15) NOT NULL
12 );
```

Listing 1: Database and Table Creation

6 server.js Code

```
1 require('dotenv').config();
2 const express = require('express');
3 const mysql = require('mysql2');
4 const bodyParser = require('body-parser');
5
6 const app = express();
7 app.use(bodyParser.urlencoded({ extended: true }));
8 app.set('view engine', 'ejs');
9 app.use(express.static('public'));
10
11 const db = mysql.createConnection({
12   host: 'localhost',
13   user: 'root',
14   password: 'yourpassword',
15   database: 'studentdb'
16 });
17
18 db.connect((err) => {
19   if (err) throw err;
20   console.log("MySQL Connected...");
```

```

21 });
22
23 app.get('/', (req, res) => {
24     res.render('register');
25 });
26
27 app.post('/register', (req, res) => {
28     const { name, email, dob, department, phone } = req.body;
29     const sql = "INSERT INTO students (name, email, dob, department, phone) VALUES (?, ?, ?,
30         ?, ?)";
31     db.query(sql, [name, email, dob, department, phone], (err, result) => {
32         if (err) throw err;
33         res.redirect('/students');
34     });
35 });
36
37 app.get('/students', (req, res) => {
38     db.query("SELECT * FROM students", (err, results) => {
39         if (err) throw err;
40         res.render('students', { students: results });
41     });
42 });
43
44 app.listen(3000, () => {
45     console.log("Server running on port 3000");
46 });

```

Listing 2: Server Configuration and Routes

7 register.ejs Code

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4     <title>Student Registration</title>
5     <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel
6         ="stylesheet">
7 </head>
8 <body class="bg-light">
9     <div class="container mt-5">
10         <div class="card p-4 shadow">
11             <h2 class="text-center">Student Registration Form</h2>
12             <form action="/register" method="POST">
13                 <input type="text" name="name" class="form-control mb-3" placeholder="Full
14                 Name" required>
15                 <input type="email" name="email" class="form-control mb-3" placeholder="
16                 Email" required>
17                 <input type="date" name="dob" class="form-control mb-3" required>
18                 <select name="department" class="form-select mb-3" required>
19                     <option value="">Select Department</option>
20                     <option>CSE</option>
21                     <option>ISE</option>
22                     <option>ECE</option>
23                     <option>ME</option>
24                 </select>
25                 <input type="tel" name="phone" class="form-control mb-3" placeholder="Phone"
26                 required>
27                 <button type="submit" class="btn btn-success w-100">Register</button>
28             </form>
29         </div>
30     </div>
31 </body>
32 </html>

```

Listing 3: Student Registration Form

8 students.ejs Code

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Students List</title>
5   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel
     ="stylesheet">
6 </head>
7 <body class="bg-light">
8   <div class="container mt-5">
9     <div class="card p-4 shadow">
10      <h2 class="text-center">Registered Students</h2>
11      <table class="table table-bordered">
12        <tr>
13          <th>ID</th>
14          <th>Name</th>
15          <th>Email</th>
16          <th>DOB</th>
17          <th>Department</th>
18          <th>Phone</th>
19        </tr>
20        <% students.forEach(student => { %>
21          <tr>
22            <td><%= student.id %></td>
23            <td><%= student.name %></td>
24            <td><%= student.email %></td>
25            <td><%= student.dob.toISOString().split('T')[0] %></td>
26            <td><%= student.department %></td>
27            <td><%= student.phone %></td>
28          </tr>
29          <% }) %>
30        </table>
31      </div>
32    </div>
33 </body>
34 </html>
```

Listing 4: Registered Students Display

9 Steps After Execution

1. Run the server using: `node server.js`
2. Open browser and type: `http://localhost:3000`
3. Fill the registration form and submit.
4. Data gets stored in MySQL database.
5. Students list page displays all registered records.
6. Verify stored data using `SELECT * FROM students;` in MySQL Workbench.

Student Registration Form

Full Name
hemanth

Email
vtu3039@gmail.com

Date of Birth
20-12-2005

Department
ISE

Phone Number
9874563211

Register

Figure 1: register Dashboard

Registered Students

ID	Name	Email	DOB	Department	Phone
1	HEMACHANDRA	vtu30309@gmail.com	2016-01-31	CSE	+91-1234567890
2	priya	priya@gmail.com	2007-02-13	CSE	+91-1234567899
3	Aarav Sharma	aarav1@gmail.com	2002-01-14	CSE	9876543201
4	Vivaan Patel	vivaan2@gmail.com	2001-03-21	ISE	9876543202
5	Aditya Rao	aditya3@gmail.com	2002-05-09	ECE	9876543203
6	Krishna Reddy	krishna4@gmail.com	2001-07-18	ME	9876543204
7	Arjun Nair	arjun5@gmail.com	2002-02-10	CSE	9876543205
8	Sai Kumar	sai6@gmail.com	2001-09-24	ISE	9876543206
9	Rohan Das	rohan7@gmail.com	2002-04-16	ECE	9876543207
10	Karthik Gowda	karthik8@gmail.com	2001-12-29	ME	9876543208
11	Rahul Verma	rahul9@gmail.com	2002-06-08	CSE	9876543209
12	Manish Singh	manish10@gmail.com	2001-08-13	ISE	9876543210
13	Sneha Kulkarni	sneha11@gmail.com	2002-01-19	ECE	9876543211
14	Ankur Shetty	ankur12@gmail.com	2001-05-04	ME	9876543212

Figure 2: students Dashboard

10 Conclusion

The Student Registration System successfully demonstrates full stack web development including frontend form design, backend processing, database connectivity, data insertion, and retrieval using SELECT query.